BIRRU VAMSHI

Backend Software Engineer | Distributed Systems | Java | GCP

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Experience

FLIPKART **September 2024 – Present**

Software Development Engineer - 1

- **Built job orchestration service:** Owned the end-to-end architecture and design decisions for a distributed job orchestration service built from scratch — defining the system's execution model, failure handling strategy, and scalability approach; the service accepts requests from upstream systems, schedules Java Spark workloads on GCP Dataproc, and supports 1,000+ job executions/day with reliable asynchronous large-scale data processing.
- **Implemented fault-tolerant reconciler:** Developed a periodic reconciler (10-second cadence) to continuously scan and re-dispatch stalled or missed jobs, improving job pickup reliability under high-concurrency workloads and ensuring near real-time execution guarantees.
- **Engineered real-time operational alerting:** Designed and deployed an event-driven notification system using Kafka to alert stakeholders on validation failures in real time — eliminating the need for manual UI polling every 2 minutes, cutting Time-to-Resolution (TTR) significantly, and reducing manual monitoring effort by 100% across 3 stakeholder teams.
- **Eliminated VM crash incidents via log rotation:** Introduced automated log rotation across all Azkaban, Mesos, and application VMs — previously, unmanaged log growth filled VM disks causing crashes and requiring 1 hour of manual developer intervention per incident; now fully automated with zero developer overhead.
- **Productionized and scaled service infrastructure:** Containerized backend services using Docker and deployed on Kubernetes; integrated ZooKeeper for distributed service discovery and configured Nagios for real-time system observability and failure alerting.

Projects

ResumeTailor – AI Resume Tailoring Platform | *Java 17, Spring Boot 3, PostgreSQL, Flyway, Docker* | Live Project

- Built and deployed a full-stack platform on Render that tailors a master LaTeX resume to a job description, with a single OpenAI-compatible client supporting 5 LLM providers (OpenAI, Gemini, Claude, Groq, Ollama).
- Designed a self-healing generation loop: compiles with `pdflatex`, feeds compiler errors back to the model for automatic repair, and trims or expands content (page overflow, or fill below 93%) to guarantee a single-page PDF.
- Secured the pipeline against code injection by blocking TeX primitives (\`write18`, `--shell-escape`, \`openout`) before compilation; added AES-encrypted API keys at rest, stateless JWT auth and BCrypt hashing, backed by 33 JUnit 5/Mockito tests.
- Optimized for a 512 MB RAM container: capped the HikariCP pool at 3 connections, used `JdbcTemplate` with a background queue for async visitor tracking with cached geolocation lookups, Flyway migrations, and a scheduled 90-day data purge.

Technical Skills

- **Programming Languages**: Java, C++, SQL
- **Databases**: MySQL, PostgreSQL, MongoDB, Redis
- **Backend, Distributed Systems & Messaging:** REST APIs, Spring boot, Dropwizard, Hibernate, Microservices, ZooKeeper, Kafka, Kubernetes, Docker, Apache Iceberg, Apache Spark
- **Cloud & DevOps**: Google Cloud Platform (GCP), Dataproc, Nagios
- **CS Fundamentals**: Data Structures and Algorithms, OOP, OS, DBMS

Education

Bachelor of Technology, National Institute of Technology, Raipur 2020 - 2024
Branch: Information Technology
G.P.A: 8.76/10

Achievements

- Leetcode **Knight** – Max rating **2008**, Top 3% globally
- Secured Global rank **179** in LeetCode Biweekly (29,000 participants) and **1069** in Weekly contest (25,000 participants).

Profile Links

Leetcode | GeeksforGeeks | Coding Ninjas | Codeforces